

Justin Lee

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EDUCATION

Boston College

Bachelor of Arts in Computer Science

Chestnut Hill, MA

September 2021 – May 2025

EXPERIENCE

Technical Services Engineer

Nov. 2025 – Present

Rokt (mParticle)

New York, NY

- Led end-to-end development of an Admin UI Event Cleanup Tool, designing and implementing a scalable internal UI and backend workflow to automate data cleanup operations, eliminating manual SQL for **5+** engineers and **15+** account managers and saving **10+** hours per week
- Diagnosed and resolved **75+** high-priority data integrity and pipeline incidents across the mParticle CDP (data pipelines, warehouse sync failures, data loss, and validation errors) by debugging Airflow ETL pipelines, inspecting S3 objects, and querying warehouse tables in BigQuery and Redshift
- Owned P1 incidents (e.g., NBCU) end-to-end, leading root cause analysis across logs, metrics, and DynamoDB identity systems, coordinating with multiple engineering teams, and delivering proactive stakeholder communication throughout incident lifecycles

Backend Software Engineer Intern

June 2024 – August 2024

Teamworks

Durham, NC

- Engineered a Python/Flask data export pipeline producing PDF/CSV outputs for **1000+** multi-team trip itineraries, automating extraction and transformation of scheduling data and eliminating manual reporting work.
- Designed and shipped core data models and backend API endpoints for a Facility Management system now used by **6,300+**, structuring relational schemas for consistent facility-data referencing across calendars, messaging, and booking modules
- Boosted code coverage by **10%** through test-driven development (TDD), writing thorough unit tests & snapshot tests that increased deployment confidence and caught edge case regressions pre-release.

Software Engineer Intern

June 2023 – August 2023

LG Electronics

Englewood Cliffs, NJ

- Delivered **10%** improvement in logistics accuracy by building a Flask-based shipment validation tool that cross-referenced customer addresses using Google Maps API, preventing delivery errors and reducing processing time for **5+** warehouse teams.
- Improved data visibility for **50+** sales reps by automating real-time synchronization between MSSQL and Airtable using Python (pyodbc) and Airtable's API, streamlining data access across product and logistics teams.

PROJECTS

Content Academy | *React, Node.js, PostgreSQL*

- Led a team of 4 engineers and designers to architect and deploy a scalable full-stack SaaS platform using React, Node.js, Redis, and PostgreSQL, supporting **750+** active users and integrating third-party APIs for personalized video content and displaying user analytics.
- Reduced dashboard and leaderboard load times by over **75%** (4s → 981ms) by implementing Redis caching, reducing redundant database reads and dramatically improving API response times in production.

Resolve | *Flask, Next.js, OpenAI Whisper, PostgreSQL, Google Cloud Platform*

- Built full-stack multimodal AI platform to analyze customer complaints across audio, text, image and video inputs, enabling real-time categorization and summarization of the complaints.
- Reduced manual customer support requests by **25%** by integrating a RAG pipeline, enabling customers to source & query data directly from our dashboard.
- Containerized the application with Docker and deployed on Google Cloud Platform, ensuring scalability and reliability.

TECHNICAL SKILLS

Languages: Python, Javascript, Typescript, C++, SQL (PostgreSQL, MySQL), HTML/CSS

Frameworks & Libraries: React, Node.js, Flask, Next.js, TailwindCSS, LangChain, OpenAI API, SQLAlchemy

Developer Tools: Git, AWS, Docker, Redis, Kubernetes, Postman, Pytest, Pinecone, Google Cloud Platform

Key Skills: RESTful APIs, JWT Authentication, RAG pipelines, Vector Database, Agile Development, Unit Testing